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Harrod Text 2/06

- Fix graphs (z ↓ in Fig 2) ✓

-

✓ - End: if $g_y^w = g_{L_D}^*$ (desired growth rate of employment), then

$$\left[s_p + \theta(1-s_p) + \mu - \frac{G_t X_t}{Y_t}\right] R = g_{L_D}^*$$

→ complex adjustments

pital
54
UK

MA-10-45

LECTURE OUTLINE Corrections 10/17/05

Economic Policy in a Growth Context: A Synthesis of Keynes and Harrod

Anwar Shaikh

Graduate Faculty
Department of Economics
New School for Social Research
9/12/05 (v.3)

~~Abstract~~ SR Equil: $I = S = sY \rightarrow Y = \bar{I}/s$

1) Standard Keynesian theory: in the SR, ~~an increase in exogenous demand~~ (investment, ~~government spending and exports) will always induce an increase in output~~. mvmt in exog, so $Y\uparrow$ as $I\uparrow$
   - This is the multiplier.
   - Modern Keynesians extend to the LR by allowing for growth in the elements of exogenous demand. $\boxed{Y = \frac{I+G+X}{s}, \quad s = s_{PRW}(1-\theta)+\theta+\mu}$
   - This introduces some new aspects, but leaves the basic theory unchanged.
2) Harrod adopts the Keynesian short run framework as his own starting point, but argues that in the long run investment cannot be treated as exogenous:
   - (1) investment not only induces production through the multiplier, but also simultaneously expands capacity. → see (note) on capacity (p 2)
   - (2) show that investment is sustainable only if it *self-consistent,* and for this to hold it must follow a *particular* growth path which he calls the warranted path. : Investment motivated by profit on desired rate of utilizn of new capital $I = sY$ → this defines Y_p
   - (3) NOT SAY'S LAW: BOTH DD & SS ADJUST
3) This path can now be shown to be very stable (proofs in the Appendices of the paper)
4) Thus the fundamental difference between Keynesian and Harrodian analysis of the long run turns on the *treatment of investment.*
5) three distinct types of contrasts. in more general case
   - First, output may or may not grow in Keynesian theory, depending on the paths of the components of exogenous demand; whereas growth is normal in Harrodian theory, because investment is only sustainable when it is growing.
   - Second, a fall in the savings rate, or in the tax rate or import propensity will raise output in the Keynesian system without necessarily harming growth; but in the Harrodian system, these same changes would lower the long run rate of growth.
   - Finally, in KK theory increases in levels and/or growth rates G, X will unambiguously stimulate output; in the Harrodian, these will only stimulate output if (G+X)/Y does not rise ➔ if (G+X) does not rise faster than existing ~~growth rate~~ warranted of output (Appendix B)

$I = sY - (G+X)$ (see above) → $\boxed{g_K = sRu_K}$ → $Y \cong Y_p \rightarrow u = 1$ → $\boxed{g_K^W = sR}$

$g_K = sRu_K - (G+X)/Y$

$g_K^W = (g_Y^W) = sR - (G+X)/Y$

Possibility of too much of a good thing: If level of G↑, then level of Y↑. But if growth of $(G+X)_t > (g_Y^W)_t$, then $(G+X)/Y \uparrow \rightarrow g_Y^W \downarrow$

GRAPHS

Economic Policy in a Growth Context: A Synthesis of Keynes and Harrod

- u – 1 is the key signal to investment
  - Not sufficient to say that I = f(u, …) (Taylor, Lavoie)
  - For Harrodian argument I = f(u-1) in such a way as to make u ≈ 1 (proofs in Appendix)
- ~~normal capacity ≈ demand (u = 1) does not imply Say's Law~~
  - ~~Say's Law says any supply will generate a matching demand~~
  - ~~classical and Harrodian notion that u ≈ 1 is based on *mutual* adj of capacity and DD~~
  - ~~This scissors has two blades.~~

Note

- normal capacity output ≠ engineering capacity, which is the *maximum* sustainable output of a given structure of plant and equipment.
- Actual output < engineering capacity, depending on shifts, overtime, intensity
- unit costs vary with degree of utilization ➔ lowest cost point defines normal capacity (Kurz)
- perfectly compatible with desired reserve capacity (like Md can include desired cash reserves)
- hence true excess capacity is when u < 1, where u* = 1 includes desired reserves.

3. Aggregate Demand and the multiplier

≡ (C+S)+M

Z = demand, Y = supply, I = investment, S = savings, G = government spending, T = taxes, X = exports, and IM = imports.

C+ C+

1) $E \equiv Z - Y = (I + G + X) - (S + T + IM) = (I - S) + (G - T) + (X - IM)$

Assume leakages proportional to income: $S = s_{priv}\cdot(Y-T)$, $T = \theta\cdot Y$, $IM = \mu\cdot Y$ ➔ $s = s_{priv} + \theta\cdot(1 - s_{priv}) + \mu$

2) $Y = (I + G + X)/s$

Growth lecture 5/25/06

1. Harrod's theory of firm → $u \cong u^*$
   → see 5/23/06, p. 6
   → problem of aggregate growth

2. Harrod's formulation of the problem
   (1) Pictures of Growth → DATA
   (b) Problem of u_K

3. Implications for PK theory

4. The problem from a Marxian & Keynesian point of view

   → solution: $s_B \equiv r$ is endogenous
   → "freeze" ~~invest~~ g_K to be a function of $(r-i)$

5. LR

$$u_L \cong \text{const} \Leftrightarrow g_K \cong g_y + g_N$$

(1) But if $g_K = f(r-i)$, then how is this to be maintained

Pasinetti-Kaldor
r adjusts
s/v is endog

Marx
g_y adjusts

Economic Policy in a Growth Context

C. Harrod-Godley Synthesis

New Cambridge view of Godley: if there were *exact twin deficits*, G-T and X-M would exactly offset each other (see three balances)

$$(I - S) + (G - T) + (X - IM) \equiv 0 \rightarrow (I - S) = 0$$
$$\rightarrow g_{Yt}^{w} = [s_p \cdot (1-\theta)] \cdot R$$

- Budget deficit would <u>not</u> affect warranted rate of growth
- Budget deficit would only worsen current account
  → *New Cambridge external limit to fiscal policy*
- Long run growth still rises if private savings rate rises (and to lesser extent, if tax rate *falls*)

If there were only "sibling" deficits, then fiscal policy is partially limited by Harrodian internal limit and partially by Godleyian external limit.

- *Harrod, Keynes and Godley all believe that state can mitigate unemployment*

A. Shaikh

Economic Policy in a Growth Context

Concluding Remarks

i. Robinson's objection

In Keynes (and in Marx), $I_t = f(r_t^e - i) = f$(expected long term profit rate of enterprise) ➔ profitability drives accumulation

Yet in Harrod, capital and investment must grow at the warranted rate ($s \cdot R$), which is not related to the expected rate of profit on investment

- How to retain Marx-Keynes view of profit-driven investment and also have Marx-Harrod self-consistent investment?

- In my view, solution lies in theory of business savings ➔ the overall savings rate is *endogenous*

ii. Employment Issue

Warranted path is given by ($s \cdot R$) ➔ unemployment rate may be rising or falling without limit (Harrod's natural rate problem)

- Solow's solution: *endogenize R* via aggregate production function
- Romer/Lucas Solution: *endogenize R* via endogenous technical change (AK model)
- Pasinetti-Kaldor-Goodwin solution: make the savings rate a function of wage-profit split, and make wage share adjust to make warranted rate = natural rate ➔ *endogenize wage share*
    - *Note that this leaves very limited room for "class struggle"*
- Marx's RAL argument: how to account for social and historical effects on wage share, and still have bounded rate of unemployment? ➔ endogenize mechanization, which responds not only to profitability but also to scarcity of labor (size of RAL)